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RE2: End-to-End Machine Learning Project – Guided Hands-On Review & Reflection

The main machine learning (ML) problem addressed in Chapter 2 of the textbook is building a model to predict housing prices in California using census data. While the problem can change based on what is the goal and perspective, the underlying goal of this project is to use the housing prices data to estimate a district's median housing price. Based on the problem and data being used, this task is a supervised regression problem.

The example the textbook uses goes over the main steps of what an end-to-end workflow would be like for ML. In short, the workflow can be broken down into 5 steps, starting with...

1. Problem Framing

Before starting the ML process, we have to know what ML problem we are trying to solve. What is the goal, benefit, and objective of doing ML in the first place? What are the algorithms, performance metrics, and data you will use?

2. Data Loading and Exploration

The next step is to start looking into the data you are going to use. Load the data as a `DataFrame`, and use exploration and visualization functions like...

- `head()` – to see the first few lines of data you are working with
- `value_counts()` – to expand on specific feature object types and get their value counts.
- `describe()` – show a summary of numerical attributes
- `hist()` – create histogram plots for each numerical attribute
- `corr()` – get the standard correlation coefficient between every feature pair

There are many more ways to explore the data, depending on the data used, but this step in exploring the data helps get an idea of what data we are working with.

3. Data preprocessing

Once we're done with the exploration and have gotten an idea of the data we are working with, we can get started with splitting the data into training and test sets, and cleaning the data of unusable data like NAs and changing categorical data into usable numerical data, standardizing values.

4. Model training and Evaluation

At this point, the data is ready to be used for model training, and we would use different training models like `LinearRegression()`, `cross_val_score()`, and `RandomForestRegressor()`. It's important to train and compare the effectiveness of different models. Running a model, we use different evaluation metrics like Root Mean Squared Error (RMSE), which help quantify how far predictions are from the actual target values. We keep repeating this step, fine-tuning the model by running the model on the test set until we are satisfied with the model's performance in prediction

I have run ML code before, so there is not much to learn from this other than to have a reference of what the actual code syntax should be. But reading the textbook is not fruitless. There were two takeaways: knowing the proper pipeline for doing ML and how to fine-tune a model. After finishing this chapter, it allows me to be prepared for what to expect from the upcoming quarter-long ML team project. While I know I will not be using the same housing data as the textbook, with a task list outlining what should be done before, during, and after training my model, this can help me avoid getting lost in what I should do depending on the situation my model might encounter. For example, if my model is underfitting or overfitting, I can refer back to the textbook to see what it did and apply similar logic and techniques to address my own problems. I feel confident that I will be able to keep track of what I need to do in case I get lost